

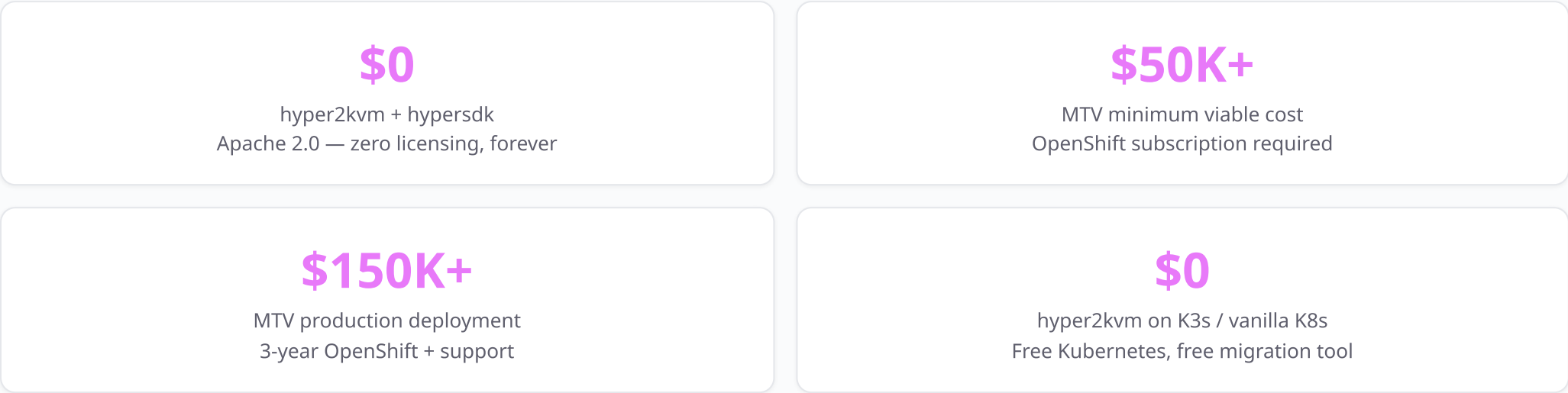
hyper2kvm + hypersdk vs Red Hat MTV

Why the Open Platform Beats the Walled Garden

A comprehensive technical and financial comparison. MTV locks you into OpenShift at \$50K+/year. hyper2kvm + hypersdk deploys to any KVM host, any Kubernetes cluster, any cloud — for \$0. Here's every advantage, feature by feature.

The \$50K Question

MTV is "free" — but only if you already pay for OpenShift. Here's the real cost.



Total Cost of Ownership — 3 Year Comparison

Cost Item	hyper2kvm + hypersdk	Red Hat MTV
Migration tool license	\$0 (Apache 2.0)	\$0 (bundled with OpenShift)
Platform requirement	Any Linux (\$0)	OpenShift (\$50K-200K/yr)
Kubernetes (if needed)	K3s / vanilla K8s (\$0)	Must be OpenShift (\$50K+/yr)
KubeVirt runtime	Open source KubeVirt (\$0)	OpenShift Virtualization (bundled)
Training	1-2 days (CLI + TUI)	5-10 days (OpenShift + MTV + K8s)
Infrastructure	Existing Linux servers	Dedicated OpenShift cluster (3+ nodes)
Support	Community + optional commercial	Red Hat subscription
3-Year Total (100 VMs)	\$0 - \$15K	\$150K - \$600K

Key insight: MTV isn't a migration tool you buy — it's a feature of OpenShift you're forced to subscribe to. If you don't already run OpenShift, you're paying \$50K+ just to get access to a migration tool. hyper2kvm works on any Linux box you already have.

Feature Matrix

Every capability compared head-to-head.

Capability	hyper2kvm + hypersdk	Red Hat MTV
VMware vSphere export	✓ Built-in govc + VDDK	✓ vSphere provider
Hyper-V migration	✓ Native VHD/VHDX	Partial (community effort)
Cloud sources (AWS, Azure, GCP...)	✓ 10 cloud providers	× Not supported
Target: Standalone KVM/libvirt	✓ Primary target	× KubeVirt only
Target: KubeVirt	✓ Any K8s cluster	✓ OpenShift only
Target: K3s / vanilla K8s	✓ Verified on K3s, EKS	× OpenShift required
Target: OpenShift	✓ Supported	✓ Native
Interactive TUI	✓ zkvm (Bubble Tea)	× Web UI only
CLI automation	✓ h2kvmctl + YAML	kubectrl + CRDs
Offline guest OS fixes	✓ VMCraft engine	Limited (virt-v2v subset)
Windows VirtIO injection	✓ Full pipeline	✓
Boot verification	✓ Automated test	×
Live progress streaming	✓ Real-time + ETA	Web UI progress only
Pre-flight validation	✓ h2kvmctl doctor	Basic validation
Job scheduling	✓ hypersdk scheduler	×
Carbon-aware scheduling	✓ Green migration	×
Cost estimation	✓ Per-VM cost calc	×
Air-gap / offline	✓ Fully offline	Partial (needs OCP)
No libguestfs required	✓ VMCraft = Python + qemu-nbd	× Uses libguestfs

Hexagonal architecture	✓ 13+ port interfaces	× Monolithic operator
Webhook notifications	✓ Slack, Teams, custom	×
RBAC + audit trail	✓ Built-in	Inherits from OpenShift

KubeVirt Deployment — The Big Advantage

Both tools deploy to KubeVirt. But the platform requirements couldn't be more different.

hyper2kvm KubeVirt Deploy

- **Any Kubernetes:** K3s, vanilla K8s, EKS, GKE, AKS, OpenShift
- **Minimal cluster:** Single-node K3s works (tested)
- **Free runtime:** Open source KubeVirt + CDI
- **Manifest generation:** Auto-creates VirtualMachine YAML
- **Disk upload:** Automated PVC creation + image upload
- **Dual target:** Same pipeline does KVM and/or KubeVirt
- **Edge-ready:** K3s on a Raspberry Pi or edge box
- **No operator lock-in:** Standard K8s resources

MTV KubeVirt Deploy

- **OpenShift only:** Will not run on vanilla K8s or K3s
- **Full cluster required:** 3+ control plane + 3+ worker nodes
- **Paid runtime:** OpenShift Virtualization (subscription)
- **CRD-based:** MTV-specific custom resources
- **Operator dependency:** MTV operator + OCP operator lifecycle
- **No standalone KVM:** Cannot deploy to libvirt at all
- **Heavy footprint:** Not suitable for edge / small sites
- **Vendor lock:** Tied to OpenShift ecosystem

Real-World Example: 50-Site Edge Deployment

A retail chain with 50 stores needs to migrate VMware VMs to KubeVirt at each site.

Factor	hyper2kvm + K3s	MTV + OpenShift
Per-site hardware	1 server (any Linux)	6+ servers (3 control + 3 worker)
Per-site software cost	\$0	\$50K+ OpenShift sub
50-site total cost	\$0	\$2.5M+ in licenses alone
Setup time per site	30 minutes (K3s + hyper2kvm)	2-3 days (OpenShift install + MTV)
Operator expertise	Basic Linux admin	OpenShift certified admin
Air-gap support	Yes — single binary + images	Complex — mirror registry needed

Architecture Comparison

Modular platform vs monolithic operator.

hyper2kvm + hypersdk: Two-Layer Platform

- **hypersdk (Go daemon):** Cloud API abstraction, 10 providers, hexagonal architecture with 13+ port interfaces
- **h2kvmctl (Python):** Disk conversion, VMCraft guest fixes, libvirt XML generation, KubeVirt manifests
- **zkvm (Go TUI):** Interactive terminal UI, vSphere discovery, live progress, file browser
- **Clean separation:** Each layer can be used independently or together
- **Extensible:** Add a new cloud provider by implementing one Go interface
- **Lightweight:** No appliance boot, no supermin, pure Python + qemu-nbd

MTV: Kubernetes Operator

- **Single operator:** Runs inside OpenShift as a set of pods
- **CRD-driven:** Migration plans defined as custom resources
- **Depends on:** OpenShift, OLM, OpenShift Virtualization, CDI, virt-v2v
- **VMware-focused:** Primary support is vSphere, limited Hyper-V
- **No standalone use:** Cannot run outside Kubernetes/OpenShift
- **Heavy dependencies:** libguestfs appliance, virt-v2v, nbdkit stack

Provider & Source Coverage

Source Platform	hyper2kvm + hypersdk	MTV
VMware vSphere	✓ govc + VDDK + OVA/OVF	✓ vSphere provider
Hyper-V	✓ Native VHD/VHDX	Partial / community
AWS EC2	✓ EBS snapshot export	✗
Microsoft Azure	✓ Managed disk export	✗
Google Cloud	✓ Persistent disk export	✗
Oracle Cloud (OCI)	✓ Boot volume export	✗
OpenStack	✓ Glance / Cinder	Community provider
Alibaba Cloud	✓ ECS disk export	✗
Proxmox VE	✓ API export	✗

KubeVirt (cross-cluster)	✓ PVC export	×
Local disk images	✓ VMDK/VHD/RAW/OVA	× (must be on vSphere)

Enterprise Features MTV Doesn't Have

Capabilities that only hyper2kvm + hypersdk provides.

Standalone KVM Deploy

Generate optimized libvirt domain XML, virsh define, and boot test — all automated. MTV has **zero** standalone KVM capability. If you're migrating to traditional KVM/QEMU hosts, MTV simply cannot help.

Boot Verification

After conversion, hyper2kvm automatically starts the VM, verifies it boots, and reports status. MTV migrates the disk and hopes for the best — no automated boot test.

Interactive TUI (zkvm)

Terminal-based guided workflow with file browser, vSphere discovery, form fields, and live progress. MTV has a web UI — but it requires an OpenShift console and browser access.

Job Scheduling

hypersdk provides cron scheduling, time windows, priority queues, retry policies, and concurrency limits. Schedule migrations for off-peak hours. MTV has no scheduler.

Carbon-Aware Scheduling

Schedule migrations when grid carbon intensity is low. ESG compliance, sustainability reporting. No competitor — including MTV — offers this capability.

Cost Estimation

Estimate per-VM migration cost including compute, storage, and network. Budget planning before migration starts. MTV provides no cost visibility.

Webhook Notifications

Push migration events to Slack, Teams, PagerDuty, or any webhook endpoint. Track progress without watching a console. MTV has no notification system.

Cloud Repatriation

Export VMs from AWS, Azure, GCP, OCI — bring them on-prem to KVM or KubeVirt. MTV cannot touch cloud workloads. Only vSphere (and limited Hyper-V).

Prometheus Metrics

hypersdk exposes /metrics for migration throughput, job queue depth, error rates, and provider latency. Plug into existing Grafana dashboards. MTV has basic operator metrics only.

Pre-Flight Doctor

`h2kvmctl doctor` checks 9 system areas: disk space, dependencies, permissions, network, libvirt, KubeVirt readiness. MTV discovers issues at runtime.

VMCraft Engine

Pure Python + qemu-nbd for offline guest fixes. No libguestfs appliance boot (~30s overhead). Works on minimal Linux, containers, edge devices. MTV depends on libguestfs.

Dual-Target Pipeline

Same migration run can output both libvirt XML and KubeVirt manifest. Deploy the same VM to traditional KVM *and* Kubernetes simultaneously. MTV is KubeVirt-only.

Deployment Flexibility

hyper2kvm runs everywhere. MTV runs on OpenShift — period.

Deployment Model	hyper2kvm + hypersdk	MTV
Bare metal Linux	✓ Primary use case	×
Any VM (RHEL, Ubuntu, etc)	✓	×
K3s (lightweight K8s)	✓ Verified, single-node	×
Vanilla Kubernetes	✓	×
Amazon EKS	✓	×
Google GKE	✓	×
Azure AKS	✓	×
OpenShift	✓	✓ Required
Container (Podman/Docker)	✓	×
Edge / single-server	✓	× (needs 6+ node cluster)
Air-gapped environment	✓ Single binary	Complex mirror registry
CI/CD pipeline	✓ YAML + CLI	CRDs (complex)

Infrastructure Requirements

hyper2kvm Minimum Requirements

- Any Linux distribution (RHEL, Fedora, Ubuntu, Debian, SUSE...)
- Python 3.8+ and qemu-img
- Optional: libvirt for KVM deploy
- Optional: kubectl + KubeVirt for K8s deploy
- Optional: K3s for single-node KubeVirt

MTV Minimum Requirements

- OpenShift 4.x cluster (RHCOS nodes)
- 3 control plane nodes + 3 worker nodes minimum
- OpenShift Virtualization operator installed
- MTV operator from OperatorHub
- OLM (Operator Lifecycle Manager)

- Runs on your existing infrastructure

- Requires dedicated infrastructure

Migration Workflow Comparison

Migrating 100 VMware VMs to KubeVirt — step by step.

#	Step	hyper2kvm + hypersdk	MTV
0	Platform setup	Install Python + qemu-img + K3s (30 min)	Deploy OpenShift cluster (1-3 days)
1	Tool installation	<code>pip install hyper2kvm</code> (2 min)	Install MTV operator via OLM (15 min)
2	Pre-flight check	<code>h2kvmctl doctor</code> — 9 area check	No equivalent
3	Discover VMs	zkvm TUI: browse vCenter, see CPU/RAM/state	Add vSphere provider, inventory sync
4	Plan migration	Select VMs in TUI, review execution plan	Create Migration Plan CR
5	Execute	Ctrl+R — batch export + convert + deploy	Start migration plan
6	Monitor progress	Real-time streaming + ETA + speed	Web UI progress bars
7	Guest fixes	Automatic: fstab, initramfs, VirtIO, network	virt-v2v subset (limited)
8	Boot verification	Automated boot test per VM	Manual verification required
9	Also deploy to KVM?	Yes — dual target in same pipeline	Not possible — KubeVirt only
10	Notifications	Webhook to Slack/Teams on completion	Check web UI manually

Total Time to Migrate 100 VMs

hyper2kvm: 30 min setup + ~8 hours migration = **under 1 day**
MTV: 1-3 days OCP setup + 1 day MTV setup + ~10 hours migration = **3-5 days**
hyper2kvm is faster to set up, faster to run, and costs \$0 in platform fees.

Use Cases Where **hyper2kvm** Wins

Scenarios where MTV simply cannot compete.

1. VMware Exit to Standalone KVM

The most common migration target is plain KVM/libvirt — not Kubernetes. MTV **cannot deploy to standalone KVM at all**. hyper2kvm generates libvirt XML, defines the domain, and boot-tests it. This is the #1 use case MTV misses entirely.

2. Cloud Repatriation

Bring VMs home from AWS, Azure, GCP, or OCI. hypersdk exports from 10 cloud providers, h2kvmctl converts and deploys. MTV has **zero cloud source support** — it's VMware-to-OpenShift only.

3. Edge / Branch Office

50 retail stores each need 3-5 VMs migrated. hyper2kvm + K3s runs on a single server per site (\$0). MTV requires a 6+ node OpenShift cluster per site (\$50K+ per site). **\$2.5M vs \$0**.

4. Budget-Constrained Organization

Startup, government agency, or nonprofit that can't afford OpenShift. hyper2kvm is Apache 2.0 — zero cost, forever. Full enterprise features without a subscription.

5. Multi-Cloud Consolidation

VMs scattered across VMware, AWS, and Azure — consolidate to on-prem KVM. One tool (hyper2kvm + hypersdk) handles all three sources. MTV can only handle VMware.

6. Air-Gapped / Classified

Defense, intelligence, or regulated environments with no internet. hyper2kvm is a single binary + Python package. MTV requires mirroring an entire OpenShift release + operator catalog — a multi-day project.

7. CI/CD Migration Pipeline

Automate VM migration in Jenkins/GitLab CI. hyper2kvm uses YAML configs and CLI commands. MTV requires kubectl + CRDs + operator lifecycle — far more complex to automate.

8. Hybrid Deploy (KVM + KubeVirt)

Some VMs go to traditional KVM hosts, others to KubeVirt. hyper2kvm handles both from the same pipeline with `emit_domain_xml + deploy_k8s`. MTV can only do KubeVirt — the KVM VMs need a different tool.

Technical Deep Dive

Why the architecture matters for long-term success.

hypersdk: Hexagonal Architecture

Domain Layer

70+ canonical Go types: Job, Schedule, VM, Cost, Carbon, Webhook, Audit, RBAC. Zero external dependencies. Business logic is portable and testable.

Ports Layer

13+ interfaces: JobManager, ScheduleManager, CapabilityDetector, CostEstimator, CarbonTracker, AuditLogger. Clean contracts, easy to mock and test.

Adapters Layer

REST API, provider clients, SQLite store, Prometheus metrics, webhook dispatcher, secrets vault. Swap any adapter without touching business logic.

Conversion Engine Comparison

Aspect	hyper2kvm (VMCraft)	MTV (virt-v2v)
Implementation	Pure Python + qemu-nbd	OCaml + libguestfs + supermin
Startup overhead	~0s (direct mount)	~30s (appliance boot per VM)
Dependencies	Python 3, qemu-utils	libguestfs, supermin, kernel modules
Container-friendly	Yes — runs in any container	Needs /dev/kvm + kernel modules
Hardened systems	Works on SELinux enforcing	Often fails on restricted systems
Package size	~5 MB	~500 MB+ with appliance
Extensibility	Add fix in Python — easy	Modify OCaml — steep learning curve
Test suite	1,273 Python + 116 Go tests	Limited upstream tests

Quality Assurance

hyper2kvm has **1,273 Python tests** and **116 Go tests** — all passing. The codebase is actively maintained with comprehensive coverage of edge cases: VMDK descriptor parsing, multi-extent disks, UEFI/BIOS detection, Windows registry manipulation, fstab UUID conversion, and KubeVirt manifest generation.

Final Verdict

The complete picture — why hyper2kvm + hypersdk is the better choice.

Dimension	hyper2kvm + hypersdk	Red Hat MTV
Cost	FREE Apache 2.0	\$50K+/yr OpenShift required
Scope	Full migration platform	VMware-to-KubeVirt converter
Sources	VMware, Hyper-V, 10 clouds, local	VMware (limited Hyper-V)
KVM target	✓ Primary target	× Not supported
KubeVirt target	✓ Any K8s	OpenShift only
Platform lock-in	None — runs anywhere	OpenShift ecosystem
UI	CLI + TUI + YAML	Web UI (needs OCP console)
Boot verification	✓ Automated	×
Scheduling	✓ Cron, priority, carbon-aware	×
Edge / small sites	✓ Single server	× 6+ node cluster
Air-gap	✓ Simple	Complex mirror setup
Dependencies	Lightweight (Python + qemu)	Heavy (libguestfs + OCP)
Architecture	Hexagonal, 13+ ports	Monolithic operator

Bottom Line

MTV is a capable tool — *if you're already committed to OpenShift*.
For everyone else, **hyper2kvm + hypersdk** delivers more features, more flexibility,
more deployment targets, and more source platforms — at **zero cost**.

Don't pay \$150K+ for a migration tool locked to one platform.
Use the open platform that works everywhere.

hyper2kvm + hypersdk — Enterprise VM Migration Platform | Apache 2.0 Open Source | github.com/ssahani/hyper2kvm